INTERNATIONAL MONETARY FUND

RWANDA

June 27, 2017

STAFF REPORT FOR THE 2017 ARTICLE IV CONSULTATION, SEVENTH REVIEW UNDER THE POLICY SUPPORT INSTRUMENT, AND SECOND REVIEW UNDER THE STANDBY CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS UPDATE

Risk of external debt distress:	**Low**
Augmented by significant risks stemming from domestic public and/or private external debt?	**No**

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The Debt Sustainability Analysis (DSA) was prepared jointly by IMF and World Bank staff, in consultation with the authorities.

Bank/Fund assessment of Rwanda's debt sustainability analysis indicates continuation of low risk of debt distress.[1] External debt burden indicators remain below "risk" thresholds, except for a small and temporary breach in the baseline of the debt service-to-revenue ratio, and the stress test for debt service-to-exports in 2023, when the Eurobond issued in 2013 matures. Recognizing Rwanda's investment needs on the one hand and its narrow export base and import-dependent growth on the other, the authorities are closely focused on carefully choosing the highest return projects, financed under the most favorable terms. In the context of the Compact with Africa, the authorities hope to encourage more private investment, leveraging guarantee schemes from multilateral and bilateral development partners and minimizing the government's exposure to additional liabilities.[2]

[1] This debt sustainability analysis (DSA) updates the DSA analysis contained in IMF Country Report No. 16/153 (June 2016). The fiscal year for Rwanda runs from July–June; however, this DSA is prepared on a calendar year basis. The results of this DSA were discussed with the authorities and they are in broad agreement with its conclusions.

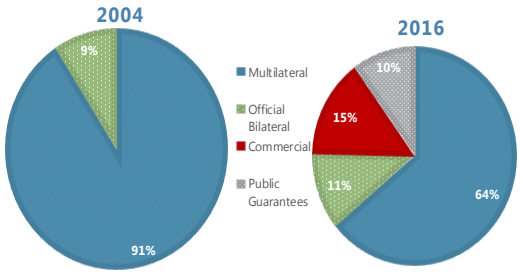
[2] Rwanda's policies and institutions are classified as "strong" under the World Bank's Country Policy and Institutional Assessment (CPIA) Index (average score in 2013-15: 3.99). The relevant indicative thresholds for this category are: 50 percent for the NPV of debt-to-GDP ratio, 200 percent for the NPV of debt-to-exports ratio, 300 percent for the NPV of debt-to-revenue ratio, 25 percent for the debt service-to-exports ratio, and 22 percent for the debt service-to-revenue ratio. These thresholds are applicable to public and publicly guaranteed external debt.

BACKGROUND

1. **Growth in the Rwandan economy decelerated in 2016.** Real GDP grew by 5.9 percent in 2016, compared to 8.9 percent in 2015, mainly due to the impact of drought on agricultural production, and to a lesser extent the completion of large investment projects in the second half of the year, and adjustment policies intended to address external imbalances. Lower commodity prices put a drag on mining exports. Despite the growth slowdown, imports increased in the first half of the year due to large public and private investment projects, causing an increase in the current account deficit. However, adjustment policies—notably sizeable exchange rate adjustment—lowered demand for imports and boosted export competitiveness in the second half of the year, such that the deterioration of the current account balance was less than forecast. In 2017, continued suppression of import demand, a levelling off in commodity prices, and robust export volume growth are projected to lead to a reduction of the current account deficit from 14.4 percent of GDP in 2016 to 10.2 percent. Real GDP growth is also projected to recover gradually, reaching 6.8 percent by 2018.

Figure 1. Public Debt
(Percent of GDP)

Figure 2. Composition of PPG External Debt
(Percent of GDP)



Sources: Rwandan authorities and IMF Staff calculations

2. **Rwanda's public sector debt has increased with an investment push in recent years, but remains comfortable in absolute terms.** At end-2016, the external debt of the public sector stood at 35.8 percent of GDP (Table 1). That ratio has increased by 14 percentage points since 2013 reflecting a sustained public investment push, including by external guaranteed debt associated with large investment projects including expansion of RwandAir and completion of the Kigali Convention Center (KCC) (Figures 1 and 2).[3] Rwanda's debt portfolio has been further affected by a shift in the composition of official development assistance away from grants toward concessional borrowing. Looking forward, a new international airport capable of handling more and larger aircraft is under construction, with the government taking a minority share in a public-private partnership. The

[3] In 2016, new debt associated with the KCC totaled US$160 million, 80 percent of which was external debt. RwandAir's continued expansion included US$171 million in loans for two new aircraft and leases for two other aircraft which, together with associated debt servicing, are included within the public sector in this analysis.

project, for which the government is expected to take on around US$37 million in external debt over 2017–19, is included in the DSA[4].

Table 1. Rwanda: External Public Debt

		2014		2015		2016		
		Billions US$	Share	Billions US$	Share	Billions US$	Share	% GDP
Multilateral creditors	0.9	1.1	60.6	1.4	61.5	1.7	60.3	21.6
Bilateral creditors	0.2	0.3	13.8	0.3	13.2	0.3	10.7	3.8
Commercial creditors	0.4	0.4	21.9	0.4	18.0	0.4	13.8	5.0
Total (excluding guarantees)	1.6	1.8	96.3	2.1	92.7	2.5	84.8	30.4
Publicly guaranteed debt	0.1	0.1	3.7	0.2	7.3	0.4	15.2	5.5
Total (including guarantees)	1.6	1.8	100.0	2.2	100.0	2.9	100.0	35.8

Source: Rwandan authorities and IMF staff.

3. **Rwanda's domestic public debt has also increased to develop a broader domestic market in recent years, but also remains low in absolute terms.** Domestic public debt was 8.6 percent of GDP at end-2016 (Table 2) close to 2 percent higher than in 2013. The increase has been driven by both short-term debt and the issuance of medium-term treasury-bonds for capital market development purposes.

UNDERLYING ASSUMPTIONS

4. **The medium and long-term macroeconomic framework underlying the DSA is consistent with the baseline scenario presented in the Staff Report for the 7th review of the PSI-supported program.** The main assumptions and projections for key macroeconomic variables are summarized in Box 1 and Table 3. The main differences between the current assumptions and those underlying the last DSA in 2015 are: i) GDP growth projections has been revised slightly down in 2018–19; ii) a slightly higher fiscal deficit is assumed due to lower projected revenues serving to increase debt and debt service measured against revenue, and iii) an improvement in the current account balance throughout the projection period, due to short and longer term adjustment policies. The reduction in external imbalances reflects, in large part, a reassessment of trade growth given stronger than expected adjustment to date—at 14.4 percent of GDP, the trade deficit was significantly lower than previously forecast in 2016, and improvements continued into Q1 2017 with contracting import volumes—particularly for consumer goods and construction goods—and robust non-traditional export volume growth, reflecting implementation of policies to encourage import substitution and promote export diversification.[5]

[4] Domestic bridge financing by the government is also included in the DSA totaling US$75 million.

[5] For instance, the authorities have launched a "Made in Rwanda" policy, to address barriers to international competitiveness with the aim of supporting domestic production (and lower imports) in key sectors including

Table 2. Rwanda: Domestic Public Debt

	2014		2015		2016		
	Billions RWF	Share	Billions RWF	Share	Billions RWF	Share	% GDP
Medium and Long-term borrowing	178	46	182	36	261	46	3.9
off which T-bonds	64		115		168		
Short-term borrowing	212	54	323	64	310	54	4.7
Total	390	100	505	100	571	100	8.6

Source: Rwandan authorities and IMF staff.

Table 3. Selected Macroeconomic Indicators, Current vs. Previous DSA

		2016	2017	2018	2019	2020
Real GDP growth (percent)	**Current DSA**	5.9	6.2	6.8	7.3	7.5
	Previous DSA	6.0	6.0	7.0	7.2	7.5
Inflation (average)	**Current DSA**	4.9	7.4	5.5	4.6	5.0
	Previous DSA	4.4	4.6	4.8	5.0	5.0
Primary balance (% of GDP)	**Current DSA**	-2.8	-2.8	-2.7	-2.7	-1.8
	Previous DSA	-3.7	-2.3	-1.9	-3.3	-2.5
Current account (% of GDP)	**Current DSA**	-14.4	-10.2	-11.2	-9.9	-8.7
	Previous DSA	-16.5	-12.0	-10.2	-11.1	-11.1
FDI (% of GDP)	**Current DSA**	2.9	3.3	4.2	4.4	4.4
	Previous DSA	3.9	4.0	4.2	4.5	4.8

Source: Rwandan authorities, IMF and World Bank staff.

construction materials, light manufacturing and agro-processing with the aim to achieve forex savings of roughly US$450 million per year.

Box 1. Macroeconomic Framework for the DSA

The medium-term and long-term framework underpinning the DSA assumes that Rwanda continues to enjoy rapid growth, with low and stable inflation.

Key highlights:

Growth: Projected long-run growth stands at 7.5 percent, unchanged from previous analysis and close to historical growth rates, and thus conservatively does not reflect a growth dividend from significant public investment in recent years. The composition of growth is anticipated to shift toward the private sector and net exports as measures designed to expand and diversify the export base and promote import substitution are assumed to be fruitful.

External Sector: Exports of goods and services (as a percent of GDP) are expected to grow consistent with historical rates reflecting, in part, strategic public investments and export promotion. Import needs are expected to remain high, although import growth rates are anticipated to be slightly below historical averages, as domestic production of certain items—such as concrete—supports import substitution. Consequently, while Rwanda's current account is projected to remain in deficit, it is expected to narrow over the period under consideration.

Inflation: Inflation is expected to remain contained. Although inflation had risen to 7.3 percent by the end of 2016, it is expected to decline to and be maintained at the authorities' medium-term target of 5 percent.

Reserves: Reserve buffers are expected to gradually increase toward 4.5 months of prospective imports, consistent with the monetary integration process among East African Community members.

Fiscal Outlook. There is assumed to be a gradual and consistent rise in domestic revenues reflecting the authorities' commitment to raise Rwanda's revenue collection efforts to a comparable level observed in other countries in the region. Primary expenditures are forecast to remain high, however, reflecting the ongoing need for significant capital and current spending.

Grants. The DSA assumes a tapering of external assistance from development partners in real terms over the projection period, reflecting reduced access to grants and greater capacity to mobilize and use domestic revenue.

External borrowing. The assumptions for new external borrowing vary over the assessment period. With the development of local bond markets and improvement in the current account position, external borrowing is expected to decline from close to 5 percent of GDP on average over the last 5 years to under 2 percent of GDP. Compositionally, from 2016-2021, the framework assumes central government external borrowing needs are met mainly by disbursements of already contracted external multilateral and bilateral debt.[1] From 2022 onward, the framework assumes that such needs will be financed with a progressively increasing share of commercial debt, including bonds issued in the international capital markets.

Domestic borrowing. The framework assumes that, over the long-term, net domestic borrowing will increase gradually from 1.4 percent of GDP on average in the last 5 years to 2.6 percent by 2037, reflecting efforts to both deepen and strengthen the domestic debt markets. Over time, the composition of that borrowing is expected to shift towards medium and long-term debt as the authorities intensify efforts to develop local government bond markets.

Domestic interest rates. New domestic borrowing is expected to be contracted at a nominal interest rate of 8 per cent—slightly below current short-term T-bill rates.

[1] Over this period, committed-but-undisbursed debt is equivalent to around 90 percent of estimated external financing needs.

DEBT SUSTAINABILITY ANALYSIS

A. External DSA

5. **Based on the assumptions outlined above, Rwanda's debt is assessed to be sustainable with low risk of debt distress (Figure 3 and Tables 4 and 5).** Like the last DSA update, Rwanda is classified as a "strong" performer, based on the quality of the country's policies and institutions as measured by the 3-year average of the ratings under the World Bank's Country Policy and Institutional Assessment (CPIA). This is reflected in higher (more accommodative) debt sustainability thresholds compared to countries operating in a weak policy environment.

6. **Under the baseline scenario, all but one debt burden indicator are projected to remain below the policy-dependent thresholds.** The only breach occurs in 2023 when the PV of debt service-to-revenue ratio just exceeds its threshold, although that breach is temporary in nature (lasting one year) and relates to when the 2013 Eurobond is set to mature. The PV of debt service-to-exports ratio also peaks in 2023, although with a small breach of the indicative threshold under the largest stress scenario—a shock to export growth. Other indicators remain well below their thresholds even under the most extreme stress scenarios. Using the probability approach, based on country-specific CPIA and historical growth information to focus on the evolution of the probability of debt distress over time, all baseline indicators remain well below their thresholds.

7. **Aside from some potential liquidity pressures when the 2013 Eurobond is set to mature, the risks to the forecast are low.** While medium term GDP assumptions are high compared to other countries, they are lower than Rwanda's historical averages: in any case, the low risk rating is robust even with somewhat lower assumptions. As the debt-service breach from the Eurobond is temporary, and considering the relatively low level of external debt, strengthening indicators of repayment capacity (the expansion of the export base and tax revenues), and that Rwanda is assumed to refinance the maturing Eurobond, also given the relatively strong capacity to develop a medium-term debt management strategy, the final assessment for Rwanda's external public and public guaranteed debt remains low risk of debt distress. However, risks have increased in recent years in line with large public investment projects. A projected continued gradual tapering of budget support and shift away from grants requires a focus on domestic revenue collection.

B. Public DSA

8. **The results of the analysis are not altered by adding domestic public debt to external debt** (see Figure 4 and Tables 6 and 7). The evolution of total public debt indicators broadly follows that of external debt under the baseline—peaking in 2019 before receding as the primary deficit begins to decline. In PV terms, debt remains significantly below the LIC DSA public debt benchmark of 74 percent for those countries with strong policies and institutions.

9. **The alternative scenarios and bounds tests indicate that the projected path for public debt indicators remain within relevant benchmarks.** Under a standard scenario that keeps the primary

balance unchanged from its 2016 level, the PV of debt-to-revenue drifts upward, highlighting the importance of securing revenue gains assumed under the baseline.

AUTHORITIES VIEWS

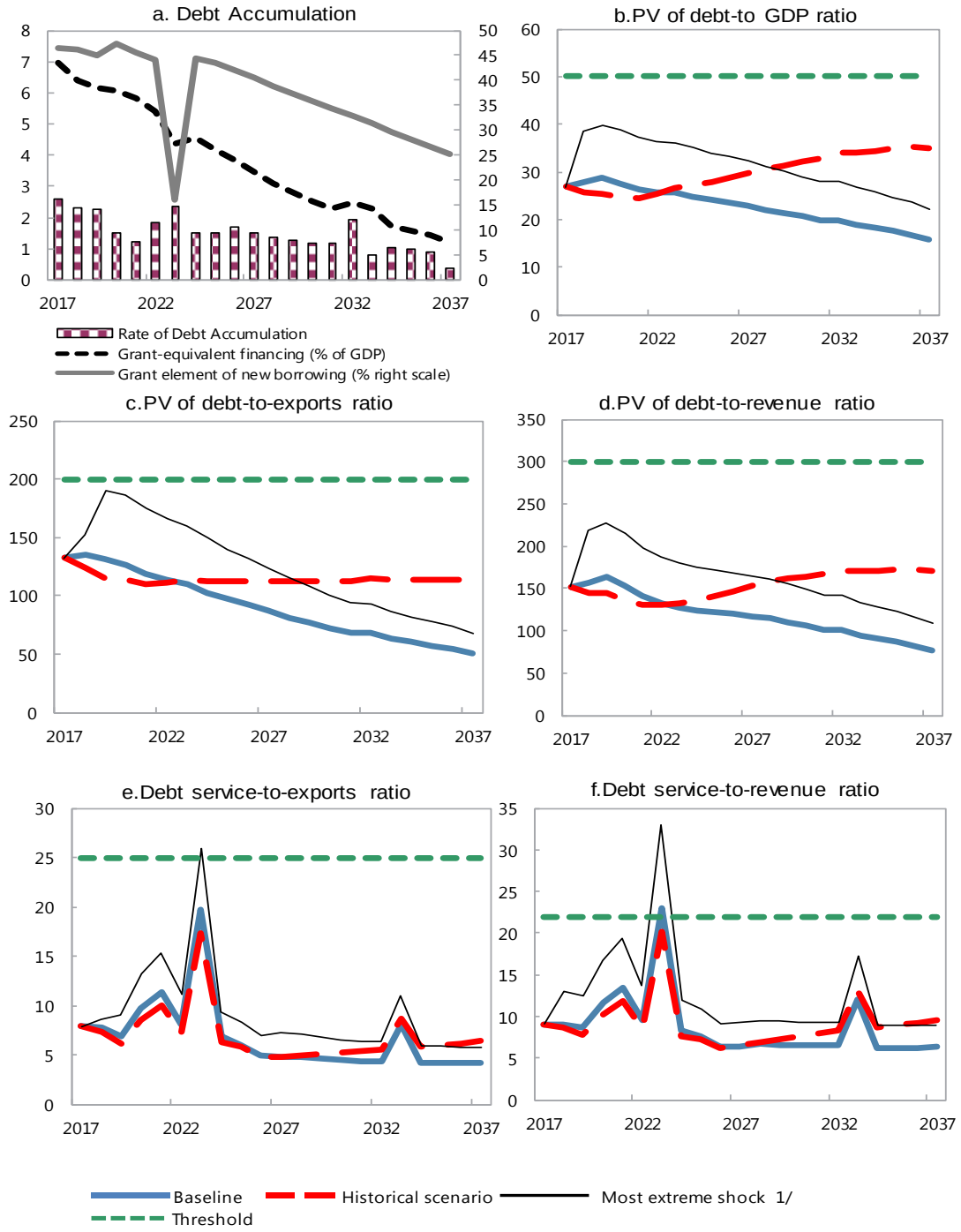
10. **The Rwandan authorities broadly agree with the results of this DSA and the overall conclusion of a low risk of external debt distress.** The authorities pay very close attention to debt sustainability, and regularly carry out their own analysis. They reiterated the commitment that their debt management strategy will be to maximize external concessional funding to avoid unsustainable debt levels, while developing the domestic capital market. The mix of domestic financing will be reoriented toward issuance of more treasury bonds vs. bills, therefore increasing the maturity length of the portfolio. They highlighted that recent and on-going investments and the implementation of measures to expand and diversify the traditional and non-traditional exports and tourism sectors should help improve resilience. The authorities also noted potential liquidity pressures when the 2013 Eurobond is set to mature, and agreed that having in place a prudent medium-term debt management strategy, and carefully prioritizing future projects and their financing are necessary to contain public debt vulnerabilities.

CONCLUSION

11. **Rwanda continues to face a low risk of debt distress.** External debt burden indicators remain below "risk" thresholds, except for a small and temporary baseline breach, in the debt service-to-revenue ratio and stress test breach of the debt service-to-exports ratio. Those breaches underscores Rwanda's susceptibility to external shocks and the potential risk of liquidity pressures in the future. However, it is judged that the risk arising from these breaches can be mitigated by the ability of the authorities to refinance non-concessional debt falling due in 2023, if sound macroeconomic and fiscal policies are maintained. Public debt, though increasing, remains comparatively low and the profile of Rwanda's external debt burden is also expected to improve over time, given expected strong growth, expansion in exports and improvement in revenues.

12. **The main risk to Rwanda's debt sustainability remains the narrow export base.** While it is assumed that this risk will be mitigated by export expansion and diversification over the assessment period, recent weakness in exports such as minerals, highlights the vulnerability that arises from a narrow export base heavily affected by fluctuating commodity prices and output. Moreover, should the anticipated medium-to longer-term expansion in exports fail to materialize, resulting in lower than expected export receipts, the risks to debt sustainability over the longer term would rise. And, more generally, while the high growth rates are expected to be sustained, policy vigilance is warranted should growth disappoint.

Figure 3. Rwanda: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2017–2037 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2027. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Table 4. Rwanda: External Debt Sustainability Framework, Baseline Scenario, 2014–2037 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2014	2015	2016			2017	2018	2019	2020	2021	2022	2017-2022 Average	2027	2037	2023-2037 Average
External debt (nominal) 1/	**27.1**	**31.1**	**39.0**			**41.3**	**43.3**	**45.2**	**44.6**	**43.7**	**43.3**		**39.2**	**29.4**	
of which: public and publicly guaranteed (PPG)	23.2	27.9	35.8			38.1	40.2	42.1	41.3	40.4	39.9		34.9	22.6	
Change in external debt	1.1	4.0	7.9			2.2	2.0	2.0	-0.6	-0.9	-0.4		-1.0	-1.5	
Identified net debt-creating flows	6.6	9.8	11.0			4.7	4.4	2.5	1.2	0.8	0.6		-0.2	-0.4	
Non-interest current account deficit	**11.1**	**12.7**	**13.7**	**8.2**	**3.8**	**9.2**	**10.3**	**9.0**	**7.8**	**7.5**	**7.2**		**6.2**	**5.6**	6.1
Deficit in balance of goods and services	16.8	17.2	18.0			12.8	13.7	12.3	11.5	10.9	10.4		8.7	6.9	
Exports	16.4	18.4	19.0			20.3	20.6	21.9	21.7	22.2	22.7		26.4	30.8	
Imports	33.2	35.6	37.0			33.1	34.3	34.2	33.3	33.2	33.2		35.1	37.7	
Net current transfers (negative = inflow)	-7.2	-6.5	-6.2	-10.0	2.6	-5.6	-5.4	-5.4	-5.2	-4.9	-4.6		-3.5	-2.0	-3.1
of which: official	-9.5	-8.4	-6.6			-6.0	-5.8	-5.7	-5.4	-5.0	-4.5		-2.9	-1.0	
Other current account flows (negative = net inflow)	1.5	2.0	2.0			2.0	2.0	2.1	1.5	1.4	1.4		1.0	0.8	
Net FDI (negative = inflow)	**-3.9**	**-2.7**	**-2.9**	**-2.5**	**0.9**	**-3.3**	**-4.2**	**-4.4**	**-4.4**	**-4.4**	**-4.4**		**-4.4**	**-4.6**	-4.5
Endogenous debt dynamics 2/	**-0.6**	**-0.2**	**0.2**			**-1.3**	**-1.7**	**-2.0**	**-2.2**	**-2.2**	**-2.2**		**-2.0**	**-1.4**	
Contribution from nominal interest rate	0.7	0.7	0.7			1.0	1.0	1.0	0.9	0.8	0.8		0.7	0.7	
Contribution from real GDP growth	-1.9	-2.3	-1.8			-2.3	-2.7	-3.0	-3.2	-3.0	-3.0		-2.7	-2.1	
Contribution from price and exchange rate changes	0.6	1.5	1.3			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-5.5**	**-5.9**	**-3.1**			**-2.4**	**-2.4**	**-0.6**	**-1.8**	**-1.7**	**-0.9**		**-0.8**	**-1.2**	-0.7
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	29.2			29.9	30.8	31.8	30.8	29.7	29.2		27.1	22.5	
In percent of exports	...	...	153.8			147.6	149.6	145.1	141.8	133.5	128.5		102.7	73.2	
PV of PPG external debt	**...**	**...**	**26.0**			**26.8**	**27.7**	**28.6**	**27.6**	**26.3**	**25.7**		**22.8**	**15.7**	
In percent of exports	**...**	**...**	**136.9**			**132.2**	**134.7**	**130.7**	**126.9**	**118.5**	**113.2**		**86.6**	**51.0**	
In percent of government revenues	**...**	**...**	**140.3**			**152.1**	**157.1**	**164.3**	**152.8**	**140.6**	**132.7**		**117.2**	**77.0**	
Debt service-to-exports ratio (in percent)	**7.8**	**6.8**	**7.0**			**10.3**	**10.2**	**9.4**	**11.8**	**13.3**	**9.9**		**6.0**	**5.2**	
PPG debt service-to-exports ratio (in percent)	**4.5**	**4.0**	**4.5**			**7.9**	**7.7**	**6.9**	**9.7**	**11.4**	**8.1**		**4.8**	**4.2**	
PPG debt service-to-revenue ratio (in percent)	**4.5**	**4.0**	**4.6**			**9.1**	**9.0**	**8.7**	**11.7**	**13.5**	**9.5**		**6.5**	**6.3**	
Total gross financing need (Billions of U.S. dollars)	0.7	1.0	1.1			0.8	0.8	0.7	0.7	0.8	0.7		0.8	1.8	
Non-interest current account deficit that stabilizes debt ratio	10.0	8.8	5.8			7.0	8.3	7.0	8.5	8.4	7.6		7.2	7.2	
Key macroeconomic assumptions															
Real GDP growth (in percent)	7.6	8.9	5.9	7.6	1.8	6.2	6.8	7.3	7.5	7.5	7.5	7.1	7.5	7.5	7.5
GDP deflator in US dollar terms (change in percent)	-2.4	-5.1	-4.1	2.7	6.6	-0.1	-1.2	-2.2	0.1	2.0	2.0	0.1	2.0	2.0	2.0
Effective interest rate (percent) 5/	2.7	2.5	2.2	1.6	0.9	2.7	2.5	2.4	2.2	2.1	2.0	2.3	2.0	2.4	2.2
Growth of exports of G&S (US dollar terms, in percent)	12.4	15.7	4.9	18.1	21.6	13.3	7.1	11.7	6.7	12.0	12.2	10.5	12.9	9.6	11.8
Growth of imports of G&S (US dollar terms, in percent)	8.9	10.9	5.3	15.6	14.8	-5.0	9.3	4.6	4.7	9.2	9.6	5.4	11.0	9.6	10.5
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	46.6	46.1	45.0	47.4	45.5	44.0	45.8	40.5	25.2	33.8
Government revenues (excluding grants, in percent of GDP)	16.5	18.4	18.5			17.6	17.6	17.4	18.0	18.7	19.4		19.5	20.4	19.9
Aid flows (in Billions of US dollars) 7/	0.9	0.9	0.8			0.7	0.7	0.7	0.7	0.7	0.7		0.8	0.6	
of which: Grants	0.6	0.5	0.4			0.4	0.4	0.4	0.4	0.4	0.5		0.5	0.5	
of which: Concessional loans	0.3	0.4	0.3			0.3	0.2	0.2	0.3	0.3	0.3		0.3	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			7.0	6.4	6.2	6.1	5.8	5.4		3.5	1.2	2.8
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			71.0	72.5	72.7	73.1	70.7	69.7		68.4	58.8	60.4
Memorandum items:															
Nominal GDP (Billions of US dollars)	8.0	8.3	8.4			8.9	9.4	9.9	10.6	11.6	12.8		20.2	50.4	
Nominal dollar GDP growth	5.1	3.3	1.6			6.1	5.6	4.8	7.6	9.6	9.6	7.2	9.6	9.6	9.6
PV of PPG external debt (in Billions of US dollars)			2.1			2.3	2.5	2.7	2.9	3.0	3.2		4.5	7.8	
(PVt-PVt-1)/GDPt-1 (in percent)						2.6	2.3	2.3	1.5	1.2	1.9	2.0	1.5	0.4	1.3
Gross workers' remittances (Billions of US dollars)	0.2	0.2	0.2			0.2	0.2	0.2	0.2	0.2	0.2		0.5	1.5	
PV of PPG external debt (in percent of GDP + remittances)	...	...	25.5			26.3	27.2	28.1	27.1	25.8	25.2		22.3	15.3	
PV of PPG external debt (in percent of exports + remittances)	...	...	123.9			120.9	123.3	120.3	116.6	109.0	104.4		79.7	46.6	
Debt service of PPG external debt (in percent of exports + remittance	...	...	4.1			7.2	7.1	6.3	8.9	10.5	7.5		4.4	3.8	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 5. Rwanda: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017–2037

(In percent)

	Projections											
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	**2027**	2037
PV of debt-to GDP ratio												
Baseline	27	28	29	28	26	26	26	25	24	23	**23**	16
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2017-2037 1/	27	26	25	25	25	25	26	27	28	29	**30**	35
A2. New public sector loans on less favorable terms in 2017-2037 2	27	28	30	30	30	30	32	32	32	31	**31**	26
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	27	27	28	28	26	26	26	25	24	24	**23**	16
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	27	28	32	32	30	29	29	28	27	26	**25**	16
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	27	28	29	28	27	26	26	25	25	24	**23**	16
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	27	27	29	28	27	26	26	25	24	24	**23**	16
B5. Combination of B1-B4 using one-half standard deviation shocks	27	26	26	26	25	24	24	23	23	22	**21**	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	27	38	40	39	37	36	36	35	34	33	**32**	22
PV of debt-to-exports ratio												
Baseline	132	135	131	127	118	113	110	103	97	92	**87**	51
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2017-2037 1/	132	124	115	113	110	111	113	113	112	113	**113**	113
A2. New public sector loans on less favorable terms in 2017-2037 2	132	136	137	140	136	134	137	132	127	123	**118**	84
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	132	130	126	125	116	111	108	101	95	90	**85**	50
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	132	152	189	187	174	166	160	149	140	132	**124**	68
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	132	130	126	125	116	111	108	101	95	90	**85**	50
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	132	132	131	129	121	115	111	104	98	93	**87**	51
B5. Combination of B1-B4 using one-half standard deviation shocks	132	126	127	125	117	112	109	102	96	91	**86**	51
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	132	130	126	125	116	111	108	101	95	90	**85**	50
PV of debt-to-revenue ratio												
Baseline	152	157	164	153	141	133	128	124	121	120	**117**	77
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2017-2037 1/	152	145	145	137	131	130	132	136	141	147	**152**	171
A2. New public sector loans on less favorable terms in 2017-2037 2	152	159	172	169	161	157	160	159	159	160	**160**	126
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	152	153	162	154	141	133	128	125	122	120	**118**	77
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	152	160	185	175	161	151	144	140	137	134	**130**	80
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	152	156	166	157	145	136	131	128	125	123	**120**	79
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	152	154	165	156	143	135	130	126	123	121	**118**	76
B5. Combination of B1-B4 using one-half standard deviation shocks	152	147	151	143	131	124	120	116	114	112	**110**	73
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	152	218	228	216	198	187	180	175	171	169	**165**	108

Table 5. Rwanda: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017–2037 (Concluded)

(In percent)

Debt service-to-exports ratio												
Baseline	8	8	7	10	11	8	20	7	6	5	**5**	4
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2017-2037 1/	8	7	6	8	10	7	17	6	6	5	**5**	6
A2. New public sector loans on less favorable terms in 2017-2037 2	8	8	7	10	12	8	20	7	7	7	**6**	6
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	8	8	7	10	11	8	20	7	6	5	**5**	4
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	8	9	9	13	15	11	26	9	8	7	**7**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	8	8	7	10	11	8	20	7	6	5	**5**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	8	8	7	10	11	8	20	7	6	5	**5**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	8	8	7	10	12	8	20	7	6	5	**5**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	8	8	7	10	11	8	20	7	6	5	**5**	4
Debt service-to-revenue ratio												
Baseline	9	9	9	12	13	10	23	8	8	6	**6**	6
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2017-2037 1/	9	9	8	10	12	8	20	8	7	6	**6**	10
A2. New public sector loans on less favorable terms in 2017-2037 2	9	9	9	12	14	10	23	9	9	8	**9**	9
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	9	9	9	12	14	10	23	8	8	6	**7**	6
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	9	9	9	12	14	10	23	9	8	7	**8**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	9	9	9	12	14	10	24	9	8	7	**7**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	9	9	9	12	14	10	23	8	8	7	**7**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	9	9	8	11	13	9	22	8	7	6	**6**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	9	13	12	17	19	14	33	12	11	9	**9**	9
Memorandum item:												
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	33	33	33	33	33	33	33	33	33	33	**33**	33

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

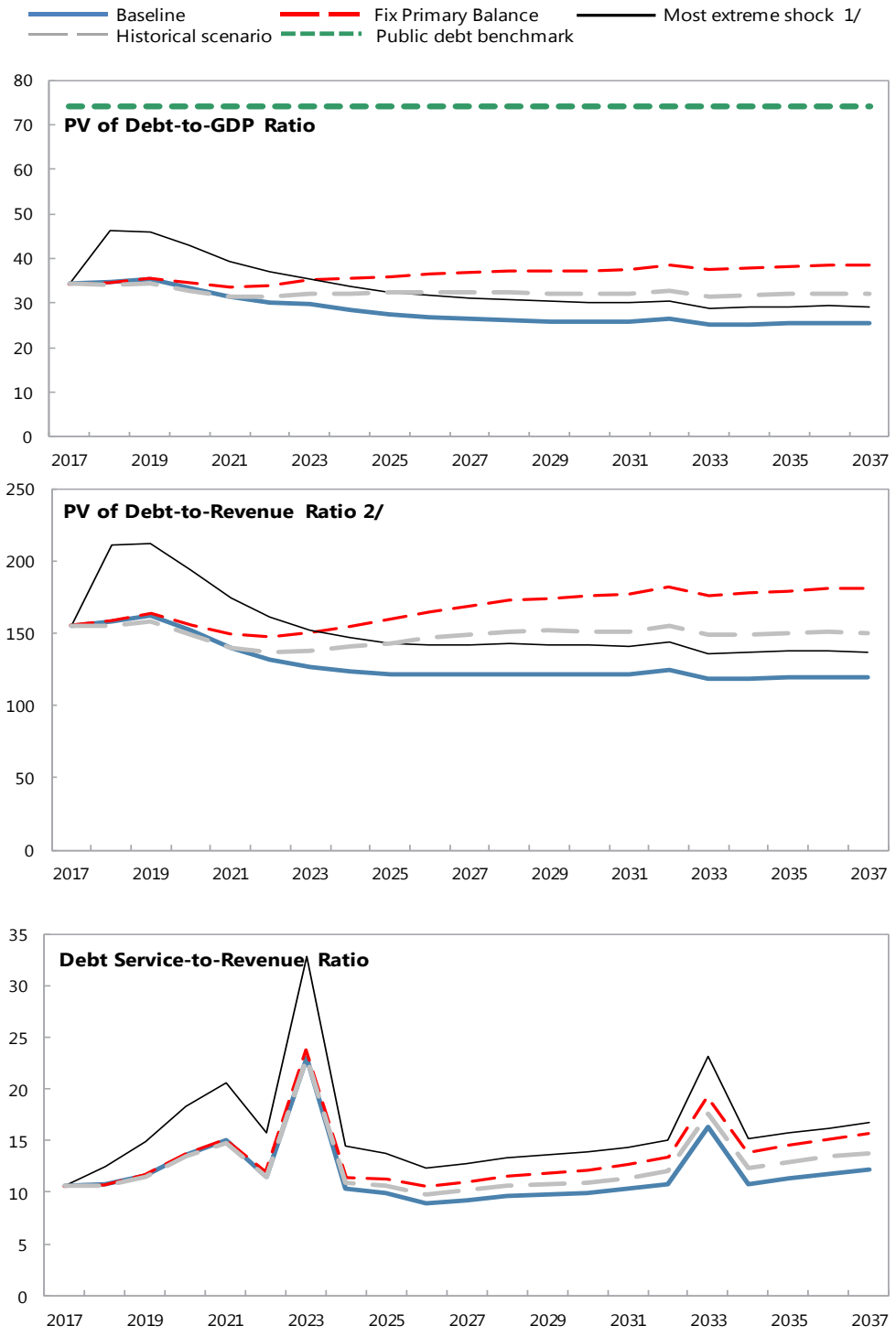
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 4. Rwanda: Indicators of Public Debt Under Alternative Scenarios, 2017–2037 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2027.
2/ Revenues are defined inclusive of grants.

Table 6. Rwanda: Public Sector Debt Sustainability Framework, Baseline Scenario, 2014–2037

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2014	2015	2016	Average 5/	Standard Deviation 5/	2017	2018	2019	2020	2021	2022	2017-22 Average	2027	2037	2023-37 Average
Public sector debt 1/	30.4	36.4	44.5			45.7	47.1	48.7	47.3	45.5	44.3		38.6	32.2	
of which: foreign-currency denominated	23.2	27.9	35.8			38.1	40.2	42.1	41.3	40.4	39.9		34.9	22.6	
Change in public sector debt	1.7	6.0	8.0			1.2	1.4	1.6	-1.4	-1.8	-1.1		-0.9	-0.9	
Identified debt-creating flows	4.3	3.4	1.7			0.5	0.7	0.9	-1.9	-2.2	-2.6		-1.6	-0.4	
Primary deficit	4.9	3.7	2.7	1.9	1.8	2.6	2.4	2.4	1.5	0.7	0.2	1.7	0.7	1.2	0.8
Revenue and grants	24.2	24.7	23.7			22.1	21.9	21.7	22.1	22.5	22.9		21.9	21.3	
of which: grants	7.7	6.3	5.1			4.5	4.3	4.3	4.1	3.8	3.6		2.4	0.9	
Primary (noninterest) expenditure	29.2	28.4	26.4			24.7	24.4	24.1	23.6	23.3	23.2		22.6	22.5	
Automatic debt dynamics	-1.3	0.0	-0.2			-2.0	-1.3	-1.1	-3.0	-2.9	-2.8		-2.3	-1.6	
Contribution from interest rate/growth differential	-2.1	-2.2	-1.8			-2.3	-2.5	-2.7	-3.0	-2.9	-2.8		-2.3	-1.2	
of which: contribution from average real interest rate	0.0	0.3	0.2			0.2	0.4	0.5	0.4	0.4	0.4		0.5	1.1	
of which: contribution from real GDP growth	-2.0	-2.5	-2.0			-2.6	-2.9	-3.2	-3.4	-3.3	-3.2		-2.8	-2.3	
Contribution from real exchange rate depreciation	0.8	2.2	1.7			0.3	1.2	1.6	0.0	0.0	0.0		...	...	
Other identified debt-creating flows	0.6	-0.4	-0.8			-0.2	-0.4	-0.4	-0.4	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.6	-0.4	-0.8			-0.2	-0.4	-0.4	-0.4	0.0	0.0		0.0	0.0	
KCC and Rwandair related						0.0									
Residual, including asset changes	-2.6	2.7	6.3			0.7	0.7	0.7	0.5	0.4	1.5		0.7	-0.4	0.5
Other Sustainability Indicators															
PV of public sector debt	...	...	34.6			34.3	34.6	35.3	33.5	31.4	30.2		26.5	25.4	
of which: foreign-currency denominated	...	...	26.0			26.8	27.7	28.6	27.6	26.3	25.7		22.8	15.7	
of which: external	...	...	26.0			26.8	27.7	28.6	27.6	26.3	25.7		22.8	15.7	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	9.8	8.6	9.1			9.1	8.5	8.6	8.5	7.9	6.3		5.2	8.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	146.4			155.2	157.6	162.5	151.5	139.4	131.5		121.4	119.0	
PV of public sector debt-to-revenue ratio (in percent)	...	...	186.9			194.8	196.1	202.3	185.6	167.9	155.7		136.2	124.5	
of which: external 3/	...	...	140.3			152.1	157.1	164.3	152.8	140.6	132.7		117.2	77.0	
Debt service-to-revenue and grants ratio (in percent) 4/	4.8	5.2	6.2			10.6	10.7	11.6	13.7	15.0	11.5		9.2	12.2	
Debt service-to-revenue ratio (in percent) 4/	7.1	7.0	7.9			13.3	13.3	14.5	16.8	18.1	13.7		10.3	12.7	
Primary deficit that stabilizes the debt-to-GDP ratio	3.2	-2.3	-5.3			1.4	1.1	0.8	3.0	2.5	1.4		1.6	2.1	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.6	8.9	5.9	7.6	1.8	6.2	6.8	7.3	7.5	7.5	7.5	7.1	7.5	7.5	7.5
Average nominal interest rate on forex debt (in percent)	2.4	2.2	2.0	2.6	3.3	2.5	2.3	2.2	2.1	1.9	1.8	2.1	1.8	1.9	1.8
Average real interest rate on domestic debt (in percent)	1.8	5.6	1.3	1.3	2.8	0.6	3.8	5.5	5.5	6.8	8.8	5.2	15.8	8.4	11.1
Real exchange rate depreciation (in percent, + indicates depreciation	3.7	10.3	6.2	0.5	6.5	1.0	3.3	4.3	-0.1	-0.1	-0.1	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.0	0.1	4.9	6.3	4.3	7.4	5.5	4.6	5.0	5.0	5.0	5.4	5.0	2.0	4.8
Growth of real primary spending (deflated by GDP deflator, in percer	9.1	6.0	-1.6	1.4	3.3	-0.5	5.3	6.2	5.3	5.7	7.0	4.8	7.3	7.5	7.3
Grant element of new external borrowing (in percent)	...	...	...	...	...	46.6	46.1	45.0	47.4	45.5	44.0	45.8	40.5	25.2	...

Sources: Country authorities; and staff estimates and projections.

1/ Refers to gross debt of the central government.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 7. Rwanda: Sensitivity Analysis for Key Indicators of Public Debt 2017–2037

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	34	35	35	33	31	30	27	25
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	34	34	34	33	31	31	32	32
A2. Primary balance is unchanged from 2017	34	35	36	34	34	34	37	39
A3. Permanently lower GDP growth 1/	34	35	36	34	32	31	30	35
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-20	34	35	36	35	33	32	29	30
B2. Primary balance is at historical average minus one standard deviations in 2018-201	34	35	37	35	33	32	28	26
B3. Combination of B1-B2 using one half standard deviation shocks	34	35	36	34	32	31	28	27
B4. One-time 30 percent real depreciation in 2018	34	46	46	43	39	37	31	29
B5. 10 percent of GDP increase in other debt-creating flows in 2018	34	41	42	40	38	36	31	27
PV of Debt-to-Revenue Ratio 2/								
Baseline	155	158	163	152	139	132	121	119
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	155	155	158	148	140	137	149	150
A2. Primary balance is unchanged from 2017	155	158	164	156	149	148	169	181
A3. Permanently lower GDP growth 1/	155	158	164	154	143	136	136	164
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-20	155	159	167	157	146	139	134	139
B2. Primary balance is at historical average minus one standard deviations in 2018-201	155	162	170	159	146	138	126	122
B3. Combination of B1-B2 using one half standard deviation shocks	155	159	166	155	143	135	126	125
B4. One-time 30 percent real depreciation in 2018	155	211	212	194	175	161	142	137
B5. 10 percent of GDP increase in other debt-creating flows in 2018	155	189	194	181	167	156	141	129
Debt Service-to-Revenue Ratio 2/								
Baseline	11	11	12	14	15	12	9	12
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	11	11	11	13	15	11	10	14
A2. Primary balance is unchanged from 2017	11	11	12	14	15	12	11	16
A3. Permanently lower GDP growth 1/	11	11	12	14	15	12	10	15
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-20	11	11	12	14	15	12	10	13
B2. Primary balance is at historical average minus one standard deviations in 2018-201	11	11	12	14	15	12	9	12
B3. Combination of B1-B2 using one half standard deviation shocks	11	11	12	14	15	12	9	13
B4. One-time 30 percent real depreciation in 2018	11	12	15	18	21	16	13	17
B5. 10 percent of GDP increase in other debt-creating flows in 2018	11	11	13	15	16	13	11	13

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.